ACADEMIC LINEAGE · ADVISOR TO ADVISOR

The Scientific Genealogy *of* Dr. Ness B. Shroff

Nineteen generations of doctoral mentorship ascending to Nicolaus Copernicus, and the scholars guided in turn.

XIX	1503	1994	51	30	1994–2026
GENERATIONS	COPERNICUS, FERRARA	SHROFF, COLUMBIA	DOCTORAL STUDENTS	POSTDOCTORAL SCHOLARS	YEARS SUPERVISING

I. THE LINE OF DESCENT

Each scholar was the doctoral advisor of the one beneath, read downward from Copernicus to Columbia in 1994. Where a scholar had two teachers the line divides and both routes are shown.

I · FERRARA, 1503

Nicolaus Copernicus — 1503
Doctorate in canon law, Ferrara · 1503 · studied Bologna, Padua, Kraków
"De revolutionibus orbium coelestium" · displaced the Earth from the centre of the cosmos
His own teachers — Leonhard von Dobschütz and Domenico Maria Novara — lie beyond this chart.

II · WITTENBERG, 1535

Georg Joachim Rheticus
Magister artium, Universität Wittenberg · 1535
The sole pupil of Copernicus; he persuaded his master to publish, and saw the book to press.
Second teacher: Johannes Volmar (Wittenberg 1515) — that line ends one step later.

III · LEIPZIG, 1567

Moritz Valentin Steinmetz
Magister artium and Dr. med., Universität Leipzig · 1550, 1567
"De Peste Capita Disputationis Ordinariae"
Second teacher: Johann Hoffmann — advisor unknown; the line ends there.

IV · LEIPZIG, 1582

Christoph Meurer
Magister artium, Universität Leipzig · 1582
"De Iride seu Arcu coelesti" — on the rainbow

V · LEIPZIG, 1604

Philipp Müller
Magister artium, Universität Leipzig · 1604
Professor of mathematics at Leipzig.

VI · LEIPZIG, 1650

Erhard Weigel
Dr. phil., Universität Leipzig · 1650
"De ascensionibus et descensionibus astronomicis dissertatio" · taught the young Leibniz at Jena

VII · ALTDORF, 1666

Gottfried Wilhelm Leibniz — 1666
Dr. jur., Universität Altdorf · 1666
"Disputatio Inauguralis de Casibus Perplexis in Jure" · co-inventor of the calculus

LEIBNIZ'S THREE OTHER TEACHERS

- **Jakob Thomasius** — Magister, Universität Leipzig · 1643
- **Bartholomäus Schwendendörffer** — Universität Leipzig
- **Christiaan Huygens** — Académie royale des sciences, Paris · 1676

The Mathematics Genealogy Project records four teachers for Leibniz. Only Weigel's line, above, climbs on to Copernicus.

VIII · PARIS

Nicolas Malebranche
Paris · philosopher and mathematician
The line divides here, and does not meet again until Poisson.

Malebranche taught two pupils; the descent runs through both, and the branches do not meet again until Poisson.

THE BASEL LINE · VIA LAGRANGE	THE PARIS LINE · VIA LAPLACE
IX · 1684 / 1687 — **Jacob Bernoulli** Universität Basel · 1684 *"Solutionem tergemini problematis arithmetici, geometrici et astronomici"*	**Pierre Varignon** Collège Mazarin, Paris · 1687 *Statics and the mechanics of forces.*
X · 1694 / 1710 — **Johann Bernoulli** Dr. med., Universität Basel · 1694 *"Dissertatio Inauguralis Physico-Anatomica de Motu Musculorum"*	**Léonor Caron** Collège Mazarin, Paris · 1710 *Taught from Varignon's own lecture notes.*
XI · 1726 / 1735 — **Leonhard Euler** Universität Basel · 1726 *"Dissertatio physica de sono"*	**Jean Le Rond d'Alembert** Collège Mazarin, Paris · 1735 *The wave equation; co-editor of the Encyclopédie.*
XII · TURIN / PARIS — **Joseph-Louis Lagrange** no formal degree · Turin and Berlin *Taught by Euler through their correspondence.*	**Pierre-Simon Laplace** Paris · 1769 *Celestial mechanics; analytic probability.*

— the two lines reunite —

XIII · PARIS, 1800

Siméon Denis Poisson
École Polytechnique, Paris · 1800
Student of both Lagrange and Laplace — the two lines reunite here.

COLLATERAL LINE · NOT ANCESTORS

- **Jean-Baptiste Joseph Fourier** — Ph.D.
- **Gustav P. Lejeune Dirichlet** — Hon., Universität Bonn · 1827
- **Rudolf O. S. Lipschitz** — Dr. phil., Universität Berlin · 1853

Fourier, Dirichlet and Lipschitz were Poisson's students — relatives of the line, not steps within it.

XIV · PARIS, 1814

Michel Chasles
École Polytechnique, Paris · 1814
Founder of modern projective geometry.

XV · PARIS, 1866

Gaston Darboux
Ph.D., École Normale Supérieure, Paris · 1866
"Sur les surfaces orthogonales"

XVI · PARIS, 1877

C. Émile (Charles) Picard
Ph.D., École Normale Supérieure, Paris · 1877
"Applications des complexes linéaires à l'étude des surfaces et des courbes gauches"

XVII · PARIS, 1926

Philippe Emmanuel Le Corbeiller
Ph.D., Université de Paris · 1926
"Contribution à l'étude des formes quadratiques à indéterminées conjuguées"

XVIII · HARVARD, 1951

Mischa Schwartz
Ph.D., Harvard University · 1951
"A Statistical Approach to the Automatic Search Problem"

XIX · COLUMBIA, 1994

Ness Bomi Shroff — 1994
Ph.D., Columbia University · October 1994
"Traffic Modeling and Analysis in High-Speed ATM Networks"

II. DOCTORAL STUDENTS · THE FIFTY-ONE

Advised to the Ph.D. at Purdue and The Ohio State University, in order of completion, 1998–2026.

1 — **Junyi Li** — PH.D. FEB. 1998
"Intercellular Coordination Schemes for Improving Spectrum Utilization in Wireless Cellular Networks"
Co-founder, Flarion Technologies; now VP of Research, Qualcomm Inc.

2 — **Jinwoo Choe** — PH.D. AUG. 1998
"Queueing Analysis of High-Speed Networks with Gaussian Traffic Models"
Associate Professor, Sogang University, Seoul
IN MEMORIAM · 2009

3 — **Paul Salama** — PH.D. AUG. 1999
"Error Concealment in Encoded Images and Video"
Professor of EE, IUPUI, Indianapolis, Indiana

4 — **Ting-Li Ling** — PH.D. DEC. 1999
"Traffic Management and Control in ATM Networks"
Senior Engineer, Onepage.com, San Jose, California

5 — **Suresh Kalyanasundaram** — PH.D. AUG. 2000
"Call Level and Class Level Quality of Service in Multiservice Networks"
Bell Labs Fellow, Nokia, Bengaluru, India

6 — **Mingbo Xiao** — PH.D. APR. 2002
"Resource Management in Power-Controlled Cellular Wireless Networks"
Professor of EE, Xiamen University, China

7 — **Xin Liu** — PH.D. AUG. 2002
"Opportunistic Scheduling in Wireless Communication Networks"
Professor of Computer Science, University of California, Davis

8 — **Han S. Kim** — PH.D. MAR. 2003
"Queueing Analysis of Loss Ratio and End-to-End Delay Distribution"
Samsung Master (Executive), Samsung Electronics, Seoul

9 — **Do Young Eun** — PH.D. MAY 2003
"Analysis, QoS Estimation, and Decomposition of Large Networks"
Professor of ECE, North Carolina State University

10 — **Dongyu Qiu** — PH.D. SEP. 2003
"Distributed Flow Control for Next Generation Networks"
Professor of EE, Concordia University, Montreal

11 — **Jang Won Lee** — PH.D. JUL. 2004
"Non-convex Optimization and Resource Allocation in Communication Networks"
Professor of EE, Yonsei University, Seoul

12 — **Xiaojun Lin** — PH.D. AUG. 2005
"Simplification of Network Dynamics in Large Systems"
Professor, School of ECE, Purdue University

13 — **Longbi Lin** — PH.D. NOV. 2006
"Energy Efficient Routing of Multi-Hop Networks"
Managing Director, Deutsche Bank, Singapore

14 — **Gaurav Sharma** — PH.D. NOV. 2006
"Design and Analysis of Wireless and Wireless-Wireline Networks"
Vice President, D. E. Shaw & Co., New York

15 — **Sungoh Kwon** — PH.D. APR. 2007
"Routing for Multihop Wireless Networks"
Professor, University of Ulsan, South Korea

16 — **Issa Khalil** — PH.D. MAY 2007
"Mitigation of Control and Data Attacks in Ad hoc and Sensor Networks"
Research Director, Qatar Computing Research Institute

17 — **Yan Wu** — PH.D. MAY 2007
"Energy Management in Sensor Networks"
Member of Technical Staff, Microsoft Inc.

18 — **Roman Chertov** — PH.D. MAY 2008
"A Device Independent Router Model"
Director of Software Engineering, Arista Networks, Santa Clara

19 — **Gagan R. Gupta** — PH.D. OCT. 2009
"Delay Efficient Control Policies for Wireless Networks"
Senior Principal Member of Technical Staff, San Ramon, California

20 — **Sarah Sellke** — PH.D. JAN. 2010
"Analytical Characterization of Internet Security Attacks"
Associate Teaching Professor of Computer Science, Northeastern University

21 — **Joohwan Kim** — PH.D. MAR. 2010
"Improving Performance of Wireless Networks Using Anycast"
Principal Engineer, Samsung Mobile, Seoul

22 — **Abdallah Khreishah** — PH.D. MAR. 2010
"Analysis and Design of Intersession Network Coding in Communication Networks"
Professor of ECE, New Jersey Institute of Technology

23 — **Sugumar Murugesan** — PH.D. AUG. 2010
"Opportunistic Scheduling Using Channel Memory in Markov-modeled Wireless Networks"
Data Scientist and Technical Lead, Johnson Controls

24 — **Srikanth Hariharan** — PH.D. DEC. 2011
"Communication-Efficient Convergecasting for Data Fusion in Wireless Sensor Networks"
Senior Manager, R&D, VMware, Palo Alto

25 — **Bo Ji** — PH.D. DEC. 2012
"Design of Efficient Resource Allocation Algorithms for Wireless Networks"
Professor of Computer Science, Virginia Tech

26 — **Shuang Li** — PH.D. MAY 2013
"Managing Dynamic Spectrum Access Under Uncertainty in Cognitive Radio Networks"
Senior Product Manager, PayPal, San Jose

27 — **Shengbo Chen** — PH.D. AUG. 2013
"Resource Allocation and Control in Communications and Cyberphysical Networks with Renewable Energy"
Professor, School of Software, Nanchang University, China

28 — **Zhoujia Mao** — PH.D. AUG. 2013
"Approximation Algorithms Design for Resource Management in Communication Networks"
Performance and Capacity Engineer, Facebook, Menlo Park

29 — **Wenzhuo Ouyang** — PH.D. APR. 2014
"Scheduling in Wireless Networks with Limited and Imperfect Information"
Manager, Amazon, Seattle

30 — **Swapna Buccapatnam** — PH.D. NOV. 2014
"Control of Large Scale Networked Systems Under Uncertainty"
Principal Inventive Scientist, AT&T Labs, Middletown, New Jersey

31 — **Yousi Zheng** — PH.D. APR. 2015
"Scheduling and Design in Cloud Computing Systems"
Senior Software Engineer, LinkedIn, San Francisco

32 — **Yang Yang** — PH.D. APR. 2015
"Throughput Characterization and Algorithmic Design in Wireless Networks"
Staff Software Engineer, Google, San Diego

33 — **Irem Eryilmaz (Koprulu)** — PH.D. MAY 2016
"Spread of Information and Influence over Dynamic Networks"
Associate Professor of Practice, The Ohio State University

34 — **Ming Zhang** — PH.D. AUG. 2018
"Game Theory and Algorithm Design in Network Security and Smart Grid"
Senior Staff Engineer and Manager, Google, Mountain View

35 — **Fei Wu** — PH.D. FEB. 2019
"Ultra-Low Delay in Complex Computing in Networked Systems"
Senior Inventive Scientist, AT&T Labs Research, New Jersey

36 — **Sinong Wang** — PH.D. FEB. 2019
"Coded Computation for Speeding up Distributed Machine Learning"
Principal Research Scientist, Facebook AI, Seattle

37 — **Zhenzhi Qian** — PH.D. APR. 2019
"Designing High-Performance Scheduling Policies in Multi-Channel Multi-antenna Networks"
Software Engineer, Google, Seattle

38 — **Jiashang Liu** — PH.D. APR. 2020
"Resource Allocation and Energy Management in Green Network Systems"
Software Engineer, BigQuery ML, Google, Kirkland

39 — **Fang Liu** — PH.D. APR. 2020
"Efficient Online Learning with Bandit Feedback"
Research Scientist, Facebook, Menlo Park

40 — **Sherif ElAzzouni** — PH.D. APR. 2020
"Algorithmic Design for Low Latency Communication in Wireless Networks"
Senior Engineer, Qualcomm, San Diego

41 — **Xingyu Zhou** — PH.D. DEC. 2020
"Load Balancing in Heavy Traffic: Theory and Algorithms"
Associate Professor of ECE, Wayne State University, Detroit

42 — **Ahmed Bedewy** — PH.D. APR. 2021
"Optimizing Data Freshness in Information Update Systems"
Senior Engineer, Qualcomm, Bridgewater, New Jersey

43 — **Wenbo Ren** — PH.D. APR. 2021
"Active Learning for Ranking from Noisy Observations"
Research Scientist, Facebook, Menlo Park

44 — **Guocong Quan** — PH.D. DEC. 2021
"Designing Caching Mechanisms for Distributed and Dynamic Caching Systems"
Assistant Professor of Computer Science and Engineering, Sun Yat-sen University, Guangzhou

45 — **Guidan Yao** — PH.D. JUN. 2022
"Information Freshness and Delay Optimization in Unreliable Wireless Systems"
Algorithm and Simulation Engineer, Nokia, Chicago

46 — **Yuntian Deng** — PH.D. JUN. 2022
"Operation of Multi-modal Transportation Networks: Optimization and Reinforcement Learning"
Applied Scientist, Amazon, Seattle

47 — **Jayanth Regatti** — PH.D. DEC. 2022
"Learning at the Edge under Resource Constraints"
Data and Applied Scientist, Microsoft Ads, Bellevue

48 — **Jiayu Pan** — PH.D. JUL. 2023
"Optimizing Age of Information and Beyond in Wireless Networks"
Assistant Professor of Software Engineering, Hangzhou, China

49 — **Sunjung Kang** — PH.D. AUG. 2024
"Enhancing Remote Tracking Systems in IoT Networks: Balancing Timely Updates and Resource Constraints"
Assistant Professor, Gwangju Institute of Science and Technology, South Korea

50 — **Hao Chen** — PH.D. JUL. 2025
"Detection, Adaptation, and Optimization of Time-Varying Stochastic Systems in Wireless Networks"
Research Scientist, Meta, Menlo Park

51 — **Yining Li** — PH.D. JUL. 2026
"Provably Efficient Algorithms for Scalable Reinforcement Learning"
Research Scientist, Meta, Menlo Park

III. POSTDOCTORAL SCHOLARS & RESEARCH ASSOCIATES

Thirty mentored since 1996. Of the twenty-eight who have moved on, twenty-five hold faculty appointments.

- **Zhenghu Gong** VISITING — 1996–97 — National University of Defense Technology, China
- **Dong Sik Kim** VISITING — 1996–99 — Professor, Hankuk University, South Korea
- **Wonjong Noh** VISITING — 2005–07 — Principal Research Engineer, Samsung Advanced Institute of Technology
- **Changhee Joo** — 2005–10 — Professor of Computer Science, Korea University
- **Jin-Ghoo Choi** — 2008–10 — Associate Professor, Yeungnam University, South Korea
- **Jian Tian** — 2009–10 — Senior Staff Engineer, Alibaba DAMO Academy
- **Yoora Kim** — 2010–13 — Assistant Professor of Mathematics, University of Ulsan
- **Zizhang Zhang** — 2010–14 — Associate Professor of Computer Science, Tulane University
- **MHR Khouzani** — 2011–13 — Senior Lecturer, Victoria University of Wellington
- **Yin Sun** — 2011–14 — Godbold Endowed Associate Professor of ECE, Auburn University
- **Jia Liu** — 2013–14 — Associate Professor of ECE, The Ohio State University
- **Joohyun Lee** — 2014–18 — Associate Professor of EE, Sogang University
- **Morteza Hashemi** — 2015–17 — Associate Professor of EECS, University of Kansas
- **Xiaowen Gong** — 2015–16 — Godbold Associate Professor of ECE, Auburn University
- **Jaewook Kwak** — 2016–18 — Software Engineer, Broadcom
- **Rahul Singh** — 2019–20 — Assistant Professor of ECE, Indian Institute of Science
- **Gourav Saha** — 2020–21 — Assistant Professor, Mahindra University, Hyderabad
- **Doyup Kim** — 2021–22 — Kyungnam University, South Korea
- **Arnob Ghosh** — 2021–23 — Assistant Professor of ECE, New Jersey Institute of Technology
- **Peizhong Ju** — 2021–24 — Assistant Professor of CS, University of Kentucky
- **Ming Shi** — 2022–24 — Assistant Professor of CS, SUNY Buffalo
- **Sen Lin** — 2022–24 — Assistant Professor of CS, University of Houston
- **Chengzhang Li** — 2023–26 — Associate Professor of CS, Shanghai Jiao Tong University
- **Mohamed A. Abd-Elmagid** — 2023–25 — Research Assistant Professor of ECE, Virginia Tech
- **Kumar Saurav** — 2024–25 — Assistant Professor of Electrical Engineering, Shiv Nadar Institution of Eminence
- **Jihyeon Yun** — 2024–26 — Assistant Professor of ECE, Kangwon National University, South Korea
- **Ziyue Luo** — 2022–26 — Associate Professor, Wuhan University, China
- **Yuchen Liang** — 2023–26 — Assistant Professor, Luddy School, Indiana University Indianapolis
- **Tasmeen Zamaan Ornee** — CURRENT — Postdoctoral scholar
- **Hongbo Li** — CURRENT — Postdoctoral scholar

Reading this chart. A scientific genealogy traces doctoral advisors rather than blood relation. Poisson studied under *both* Lagrange and Laplace, so the descent from Malebranche runs by two equal routes that reunite in 1800. The chart is capped at Copernicus because above him the record thins into mentorship reconstructed by scholars rather than degrees conferred. **Sources:** the Mathematics Genealogy Project, the original Scientific Genealogy tree, and Prof. Shroff's curriculum vitae.

LI
DOCTORATES CONFERRED